

Implementing Gesture Recognition in Your Project

Gesture recognition is very widely used in consumer devices, and some of its implementations are quite innovative. Recognising the correct gesture is the key here and this article discusses how your hardware can help you do that



What is gesture recognition

This technology enables devices to interpret human gestures using sensors, whose data is then processed using mathematical algorithms. The system is built to identify specific human gestures, and then use them either to convey information or to control a device. The gestures used have to be intuitive, simple and universally acceptable to ensure they are easily adopted by users.

Gesture recognition makes computers more accessible

for the physically-challenged and also makes interaction more natural in the 3D virtual world or in gaming. In today's world, gesture recognition is an evolving technology that can be seen all around us. The latest phones, computers, gaming consoles, and even TVs now feature it.

How it works

The human body's movements are read by an imaging sensor like a camera, which then sends that data to a computer. The computer uses the gestures captured as inputs, processes this input to understand what was

gestured, and then sends the answer as a command to the currently running application or control device.

Let us look at an example of a person clapping his hands in front of a camera. When this clapping gesture data is processed by the connected computer, it could produce the sound of cymbals, as programmed. The specialty of this technology is that users do not have to wear any special equipment or attach any device to their bodies, as is the case with some alternative technologies.

Sensing the gestures

When you gesture at your computer or mobile device, it obviously needs a way to 'see' or 'sense' what you are doing. This is where sensors come in. There are a number of sensing technologies that can be considered, but we'll stick to the popular ones here.

Accelerometers. Also known as inertial or acceleration sensors, these are used to detect the changes in force resulting from the fall, motion, tilt, shock, positioning or vibration of the body they are connected to, with respect to time. These measure linear acceleration, so if the device is in a free fall, the accelerometer will give a '0' value for acceleration.

Application. In the case of a smartphone, the direction of the screen changes as soon as we tilt or turn the phone. The direction in which we hold the phone (horizontal or vertical) is sensed by an accelerometer and the information in the form of electrical signals is sent to the screen, which automatically adjusts itself either in

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The first question that aroused my curiosity was, 'What exactly is a 'gesture'? According to a dictionary, gesture is a sequence of postures that are connected by motions over a short period of time. I did not find that really helpful. I'll explain it in an easier way—a gesture is a form of non-verbal or non-vocal communication. So your messages are communicated by visible bodily actions like waving, swiping, facial movements or even pointing to the sky.